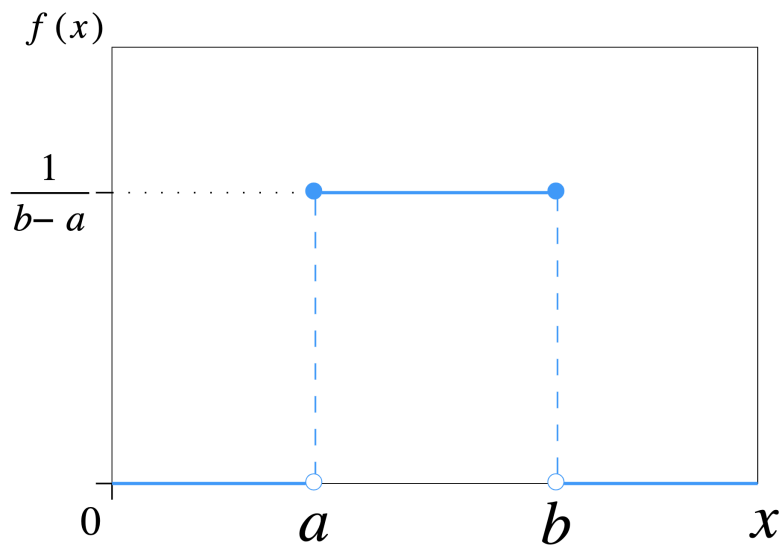


MAT 310, Handout 13: Uniform Distribution (4.4)

Uniform Distribution $X \sim U(a, b)$ is uniformly distributed over the interval (a, b) if it is equally likely to occur anytime in the interval. Its density function is given by

$$f(x) = \begin{cases} & a \leq x \leq b \\ & o.w. \end{cases}.$$

Illustration:



Example 1. Consider a bus that is equally likely to arrive at a station anytime between 2pm and 3pm. We would say that the arrival time of the bus is _____ distributed over the interval $(2, 3)$.

Example 2. Find the cdf $F(x)$ of $X \sim U(a, b)$.

Example 3. Consider the bus arriving uniformly between 2 and 3pm.

(a) What is the probability that it arrives between 2:20 and 2:45?

(b) What is the probability that it arrives before 2:50, given it hasn't arrived by 2:30?

Expected Value and Variance of $X \sim U(a, b)$

$$E(X) =$$

$$V(X) =$$

Proof:

Example 4. Consider the bus example. Find the mean and variance of the time waiting for the bus (beyond 2pm). Label the units.

Follow-Up Question: What is the standard deviation of the waiting time? Label the units.